

PROBENAV

Fast, precise and repeatable positioning of electromagnetic probes for local Side-Channel Attacks

Matthias Probst^{*}, Alexander Wiesent^{*}, Michael Gruber[†], Georg Sigl^{*†}

^{*}*School of Computation, Information and Technology, Technical University of Munich, Munich, Germany*

[†]*Fraunhofer Institute for Applied and Integrated Security (AISEC), Munich, Germany*

{matthias.probst, alexander.wiesent}@tum.de, {michael.gruber, georg.sigl}@aisec.fraunhofer.de

Abstract—Localized side-channel analysis makes it possible to evaluate only the relevant chip area by measuring near-field electromagnetic (EM) emanations. Compared to global power measurements, this can lead to more powerful attacks as the signal-to-noise ratio is higher and irrelevant circuit components are not included in the recorded measurements. Especially for profiled attacks and their reproduction, the probe position in a localized scenario is of utmost importance. Ideally a probe should be placed identically during the profiling and attack phases, as small variations can have a large impact on the success of the attack. In this work we present our methodology – PROBENAV – to accurately reposition an EM probe which is optimized for localized measurements, i.e., near-field measurements. We evaluate cross-correlation, Oriented FAST and rotated BRIEF (ORB) and particle filters to re-calibrate the coordinate system of our setup. As a result, our methodologies show that precise positioning on a STM32F303 microcontroller is possible for a profiled attack scenario with different EM probes. Furthermore, by requiring only a single trace per position, profiling is 3 times and repositioning 28 faster in terms of number of collected traces compared to the state of the art.

Index Terms—local EM-measurements, Probe Positioning, SCA

I. INTRODUCTION

Recently, physical attacks have posed an increasing threat to cryptographic implementations due to the rise of the Internet of Things (IoT). Although modern cryptographic functions are tested to be secure against cryptanalytic attacks while undergoing strict standardization, these attacks assume a black-box scenario where an attacker does not know the internals. However, since cryptographic functions need to be implemented on a physical device, access to this device is possible. This provides additional knowledge to an attacker by, for example, observing side-channel leakage, which violates the black-box assumption.

Complementary Metal Oxide Semiconductor (CMOS) based integrated circuits inherently cause data-dependent power consumption, resulting in the most commonly exploited power side-channel. Power Analysis (PA) observes the power consumption while some device executes a cryptographic algorithm. Computing intermediate values based on a key hypothesis and comparing these with multiple power measurements is the principle of Differential Power Analysis (DPA) and enables the extraction of a secret key [1]. Additionally, Electro-Magnetic (EM) measurements have been proven to work similarly and additionally provide the opportunity to observe local behavior instead of the global power consumption.

A crucial step for attacks exploiting local EM is to find a suitable position for recording them. A trivial approach is trial and error-finding positions by scanning the whole chip in a grid search and selecting the position with the best yield. Multiple works [2]–[8] aim to improve the trivial approach by using heuristic methods for maximizing the leakage while recording emanations from multiple positions.

Furthermore, machine learning is employed to precisely reposition a local EM probe in a profiled attack scenario [9]. Finding attack positions (again) in the same setup or in a profiled attack scenario is of high importance for multiple reasons: First, a security lab is typically limited by the amount of x-y-z tables and, thus, by local setups. Therefore, setups need to be adjusted to meet the needs of the current analysis to maintain a high setup utilization. Reevaluation of results is challenging even when using identical components since the same probe position cannot easily be determined by the human eye and calls for a probe positioning approach. Second, the reproduction of results on a different setup, i.e., scope or x-y-z table, etc., is also important. Additionally, other research groups may want to reproduce the results from a publication in their lab to continue research on them. For all scenarios, a robust placement strategy needs to enable the reproduction of results that are as close to the original as possible. Lastly, a different implementation on the same device can be attacked at the same or similar position, e.g., when attacking the memory bus of a microcontroller. Thus, consistently placing the probe at this position with high activity, e.g., over a memory bus or the register file, independent of the targeted algorithm is required.

In this work, we aim to simplify and speed-up the positioning solution introduced by Richter et al. [9] by replacing machine learning algorithms with techniques from image processing and navigation for enhanced robustness and explainability. Furthermore, in contrast to [9], we achieve positioning independent of the executed algorithm on the Device Under Test (DUT) as long as it is the same for profiling and attack devices¹. We show that our proposed techniques allow fast positioning in large maps with similar accuracy as in the work by Richter et al. [9]. Due to requiring less traces, we achieve a speed-up of 2.9 for profiling and 27.9 for repositioning. We call our positioning method PROBENAV since it allows us to

¹The algorithm needs to utilize the same units of the chip but does not necessarily need to execute identical code.

accurately navigate the probe over the chip.

A. Organization

The remainder of this work is structured as follows: First, we list related work and clarify our contribution to the state of the art in Sec. II. Next, we explain the principles of our positioning methods of PROBENAV in Sec. III. Our results of all three positioning strategies are outlined and compared in Sec. IV. We discuss the findings from this work in Sec. V before we conclude our work in Sec. VI.

II. RELATED WORK

The problem of either Self Localization (SL) or Simultaneous localization and mapping (SLAM) is well-studied in various research fields. Most commonly, this problem is tackled in scenarios revolving around navigation, which requires an estimation of the current position without having access to ground truth. This makes it necessary to estimate the current location by the observation of measurements that are related to the current position in conjunction with prior knowledge, i.e., maps. In recent years, the use of Global Positioning System (GPS) has become more widespread. Still, military vehicles usually cannot rely on GPS as the sole location source, as the technology is rather susceptible to jamming attempts. A possible attempt to perform SL was proposed by Irani et al. for the guidance of the BGM-109 Tomahawk cruise missile, where the localization is based on the correlation of satellite images compressed into a binary map, with images taken by an on-board camera [10]. Another more robust approach was proposed by Gustafsson et al. It uses the same principle of comparing an elevation map with radar data taken by the ground radar of a fighter jet [11]. Contrary to the previous correlation approach, this time, an approach based on a Particle Filter (PF) is used. In this work, we reuse the approaches of [10], [11] in the context of probe positioning, which is essentially the same problem under better measurement conditions without the real-time requirement. Furthermore, localization of objects within images is a well studied field in image recognition and, e.g., necessary for self driving cars or surveillance cameras. Cross-correlation similar to [11] as well as Oriented FAST and rotated BRIEF (ORB) [12] are prominent examples to fulfil such tasks and we also test their feasibility for probe-positioning in this work.

A. State of the Art

In the following, we summarize state of the art for the positioning of a probe for recording measurements of local EM-emanations within Side-Channel Analysis (SCA). Iyer et al. propose a method for position finding based on a greedy acquisition approach [6]. Their technique uses the minimum number of traces to extract a secret key from an Advanced Encryption Standard (AES) implementation as a cost function.

Danial et al. propose a low-cost framework to determine a position for EM-based SCA utilizing a gradient search algorithm [7]. The authors find the best location for a correlation-based attack by finding maximum leakage with Test Vector

Leakage Assessment (TVLA) or Signal-to-Noise Ratio (SNR). While Danial et al. aim to maximize the attack effectiveness regarding minimizing the number of required traces, other works [2]–[6] only try to identify the location with the largest information leakage without having any specific attack in mind.

In [8], the variance is analyzed with the help of the F-statistic to determine the position of an SCA-probe for SCA. The work demonstrates that some positions are ineffective and do not reveal information. However, the method is only tested at the example of AES, and identified information leakage is not evaluated in terms of being suitable for an attack.

To quickly place a probe at a suitable position, Jiang and Pavalidis propose a gradient search algorithm used to find a local maximum of the SNR [13]. The real-world applicability is evaluated in the example of memory buses, which tend to have higher EM-emanation.

Machine learning in the form of a Multi-Layer Perceptron (MLP) is also applied to place a probe at the best attack position [14]. The MLP machine learning model allows probe placement with high accuracy on the base of a single trace. The placement grid, however, needs to be trained previously.

Although multiple techniques exist to place a local EM-probe on some attack target, the reproducibility of this placement is not investigated in detail. Richter et al. [9] use a Convolutional Neural Network (CNN) to precisely reposition a near-field EM probe on a decapsulated microcontroller. For their method, they require about ten million traces for training, which also need to be preprocessed. With that, they achieve a mean positioning error of less than $50\mu\text{m}$ which on average is at around $30\mu\text{m}$. Nevertheless, the question arises whether a placement is possible independent of the implemented AES or any algorithm at all.

B. Contribution

We propose a solution for reproducible probe-positioning with a minimum amount of data collection with PROBENAV. Closest to our work is the machine-learning-based repositioning method by Richter et al. [9]. This method has a few drawbacks: First, for training the CNN ten million traces are necessary to map the area of only about 5000 measurement positions with an accuracy of $30\mu\text{m}$ on average. Mapping a larger area results in a drastic increase of training traces. Second, trace compression, i.e., knowing areas of interest in the time domain of the collected traces, is necessary to decrease the amount of data to feed into the CNN. Next, Richter et al. show the feasibility at the example of an AES implementation and are dependent on information leakage from this algorithm for accurate positioning. Lastly, the chip in their case is decapsulated, and the question arises whether the position also works with a packaged chip which features lower SNR due to the package blocking EM emanations.

With PROBENAV we aim to achieve accurate positioning with similar precision as in [9], but with less preliminaries. To reduce the traces acquisition and thereby the positioning time, we aim to use just a single trace per probe position for

profiling and testing. Additionally, the executed operation is of no importance since we aim to position based on the overall activity of the DUT². Furthermore, our results originate from a packaged chip and an area of 1 cm², where we collect over 50 times more measurement positions than Richter et al. [9] in our experiments. Even with the coverage of 50 times more positions, we gain a measuring speed-up of a factor of 2.9 for profiling and a factor 27.9 for repositioning. This is a result of PROBENAV requiring merely a single trace per position. During profiling we do not need any costly training which additionally speeds up the process compared to [9]. For comparability of results, we use an almost identical EM near-field probe, thus, the main difference lies in the depackaging of the chip, which we do not need. We additionally test the accuracy with a probe with larger coil diameter and less spacial resolution.

PROBENAV thereby is the methodology which consists of a profiling heatmap and a positioning algorithm. The positioning algorithm is essentially an image placement method and is interchangeable. Therefore, in this work, we investigate three promising choices - Cross Correlation (CC), ORB and Particle Filter (PF) - and select the most suitable for our microcontroller.

The implementation of PROBENAV is available on gitlab³ together with sample heatmaps from also evaluated in this work.

III. METHODOLOGY

To reliably reposition a local Electro-Magnetic (EM) probe, we must solve the problem of identifying the relative position of one coordinate system within another. For now, let us assume that both systems contain the same point. We call the coordinate system of later recorded points S_T while the original system is called S_I . Our goal is to find the matrix A that can transform a point p_T in the system S_T into the system S_I by computing

$$p_I = R_\varphi \cdot p_T + \begin{pmatrix} x \\ y \end{pmatrix} \quad (1)$$

where p_I is the new vector. The matrix

$$R_\varphi = \begin{bmatrix} \cos \varphi & -\sin \varphi \\ \sin \varphi & \cos \varphi \end{bmatrix} \quad (2)$$

realizes the rotation by φ in Eq. (1). Fig. 1 illustrates the problem.

In the probe positioning problem, other rotation angles can be neglected because they assume a tilted x-y-z table. Furthermore, a difference in z direction, i.e., height, can also be neglected since finding the correct height is easily possible by visual inspection with the camera and a slight variation of height had a negligible influence on the resulting heatmaps. Additionally, we noticed no change in our collected data with a slightly different height other than being attenuated.

²The operation should be the same during the profiling and testing phase for coherent results.

³gitlab.lrz.de/tueisec/probenav

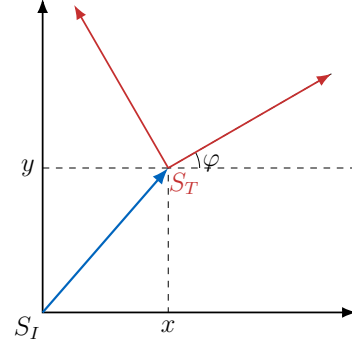


Fig. 1: Coordinate transformation problem.

Solving Eq. (1) is achieved in PROBENAV by identifying (x, y, φ) . Finding these parameters is also the solution to our probe positioning problem since the current coordinates S_T can be translated to the original heatmap with coordinates S_I , allowing us to place the probe again accurately. We aim to find the parameters with the following steps:

- 1) Profile the chip with a detailed map I (having base S_I)
- 2) *Some change in measurement setup happens ...*⁴
- 3) Recording a new map T (also called cut-out here) with a - most likely - different base S_T
- 4) Using a method to find the difference between bases (x, y, φ)
- 5) Using parameters to precisely position the probe again

Thereby, T does not need to be as large as I , as we will show in our experiments. This accelerates the repositioning drastically, since it allows for much faster mapping during the repositioning step. Furthermore, for I and T we do not need data-specific information, i.e., Test Vector Leakage Assessment (TVLA) or Correlation Power Analysis (CPA) information. We use the absolute mean, absolute maximum, absolute variance, and the Signal-to-Noise Ratio (SNR) also used in [15] as metrics to transform the traces into heatmap pixels. Those metrics require just a single trace, significantly lowering the amount of collected data. We achieved similar results with all transformations. Nevertheless, we only evaluate the results based on the heatmap computed by the variance's absolute value in this work to enable a detailed evaluation of those with reasonable computational amount.

Finding some cut-out T in some map I and thereby determining (x, y, φ) is an image recognition or navigation problem, essentially. Thus, we evaluate the applicability of imaging techniques for our problem. Our methodology PROBENAV does not require any specific placement technique. The correlation coefficient is the most common technique in this field and will be introduced in Sec. III-A. Cross Correlation (CC) is also well-known in the field of side-channel analysis and therefore an obvious choice for evaluation in this problem.

⁴Slight rotation, the shift of position, change of device (i.e., different microcontroller) or similar.

Another advanced and widely used technique from computer vision is Oriented FAST and rotated BRIEF (ORB) [12], which is described in Sec. III-B. ORB is invariant to rotation and scale and provides resilience in position finding. Furthermore, due to its design, it is less computational expensive than CC.

On the other hand, we need to move the local EM-probe to collect traces from different measurement positions. Thus, navigation techniques can provide another inspiration for solving our problem. A very robust method uses a Particle Filter (PF), which can efficiently determine a position even with a low amount of data. Thus, we explain PFs in Sec. III-C and test them in our problem. The advantage of PF is its probability based approach making it robust against noise. Additionally, a PF is not relying on image classification only but is multivariate and allows for tracking multiple attributes. We will now briefly introduce the different template matching methods for determining x, y, ϕ .

A. Cross Correlation

The cross correlation coefficient can measure the similarity of two matrices T and I . In our case, the image matrix I is always bigger than template T and T should be found within the dimensional boundaries of I .

T is matched with the normalized correlation coefficient according to

$$R(x, y) = \frac{\sum_{x', y'} \bar{T}(x', y') \cdot \bar{I}(x + x', y + y')}{\sqrt{\sum_{x', y'} \bar{T}(x', y')^2 \cdot \sum_{x', y'} \bar{I}(x + x', y + y')^2}}, \quad (3)$$

where I is the image on which we try to find the cut-out T . In Eq. (3), x', y' iterate over size of I in our case.

The result R at its respective coordinates yields a high value if I and T match more closely. Consequently, the most likely position of the center of T within I is related to the coordinates of the maximum R .

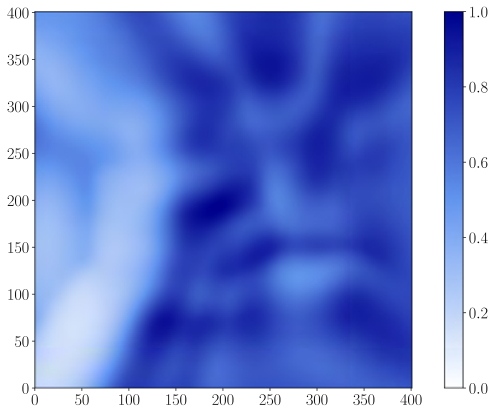


Fig. 2: Map of the correlation for each position of some cut-out T in the profiling map I . Darker colors represent higher correlation.

1) *Cross Correlation for Probe Positioning:* We apply the idea of using the cross correlation for navigation from [10] to our problem. In our case, we calculate Eq. (3) by using the

implementation from the OpenCV library [16]. Determining the position of a cut-out T on the heatmap without rotation is sufficient to make an accurate prediction. Nevertheless, we need to cover the case of a rotated cut-out since the reference setup and the rebuilt one can be slightly different in orientation. For that, we cover of $\pm 5^\circ$ in steps of 0.5° . We solve this problem algorithmically by slightly turning the cut-out over an angle of plus and minus five degrees.

Fig. 2 shows an example of the correlation results when moving T over I while the value of R is plotted at each coordinate pair. For simplicity reasons, we assume a correct rotation in Fig. 2. One can easily identify larger areas with high correlation, where the correlation values are close to each other. The reason for that is the similarities of features in these regions. Nevertheless, the absolute maximum of all correlation values constantly gives the correct position in our experiments.

B. ORB: Oriented FAST and rotated BRIEF

ORB is a computer vision solution for detecting and matching features in an image proposed by Rublee et al. [12]. It is based on the keypoint detector FAST [17] and the BRIEF descriptor [18]. ORB adds an orientation component to FAST features based on intensity centroid [19]. Further, BRIEF is slightly modified to be rotation-aware by transforming the location matrix L to a steered location matrix L_φ by

$$L_\varphi = R_\varphi L, \quad (4)$$

where R_φ is the rotation matrix by angle φ . This changes the original BRIEF classifier $f_n(\mathbf{p})$ to $g_n(\mathbf{p}, \varphi)$, which locates an image patch $\mathbf{p}$ and its angle φ based on binary features.

These methods conclude in ORB being a classifier detecting keypoints of an input. Next, features at these keypoints are extracted together with their coordinates. We benefit from the low computational complexity and the noise robustness of ORB. Another reason for choosing ORB is that it is free from licensing restrictions.

To apply ORB to image detection, we utilize the OpenCV python implementation of ORB [16]. We automatically select the settings for the search algorithm to find the optimal values for the parameters

- scale factor values $\in (1.1, 1.2, 1.3, 1.4, 1.5)$,
- number of pyramid levels $\in (4, 6, 8, 12)$,
- edge threshold $\in (3, 5, 10, 15)$,
- FAST threshold $\in (3, 5, 10, 15)$, and
- $WTA_K \in (2, 3, 4)$.

Since ORB employs a pyramid detection scheme, the scale factor gives the amount pixels in the next level. Higher scale factors reduce the amount of pyramid levels while more pixels are omitted by each level, which can degrade the feature detection capabilities. Lower factors may lead to too many pyramid levels and a high computational effort. The edge threshold specifies the size of edge features, while the FAST threshold gives the setting for finding stable keypoints. Finally, WTA_K gives the number of features for the BRIEF feature descriptor. The patch size and the score type is kept at the default (31 and Harris Score, respectively) for our experiments.

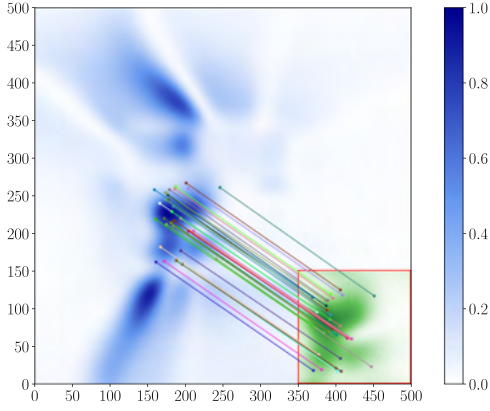


Fig. 3: Matching of keypoints in I from the green cut-out T in the bottom left. Matching keypoints are connected by the colored lines.

The selection is done on the fly while matching T in I , by counting the matches with the currently tested parameter set. The ORB-classifier is created with the parameter set. These settings extract the keypoints and descriptors from both I and T . Next, the descriptors of both are matched, and the distance between both sets is calculated. The best parameter set is recognized with the least distance in the matches. This parameter set is then used to find the position of T in I and get the resulting shift in x, y , and angle. Thereby, the positioning is determined by the highest number of matches, where the corresponding keypoints have the same distance to each other. In other words, the translation which results from the highest number of matches gives the positioning result. A position example is depicted in Fig. 3, where the matches in T and I are illustrated by connecting the corresponding keypoints.

C. Particle Filters

The following brief introduction on PF is based on the illustrated guide by Labbe [20]. Particle filters are a tool to determine the position of a moving object in space [10]. The idea behind this method is to randomly spread multiple points (particles) representing a possible system state. After transforming the points with our system, a weighting process occurs based on how closely the particle represents the observations.

Determination of positioning is achieved in an iterative process, where five steps follow each other within one iteration. First, N randomly distributed parameters are generated. These contain a position, a direction of movement, and other variable(s), that may be relevant to the problem. Furthermore, a weight indicates the likelihood of representing the actual state of the system. As a second step, the believed behavior of the actual system is applied to the particles, i.e., a movement in some direction is applied to all particles. Thirdly, measurements of real-world parameters are considered by updating the weights of all particles. This results in particles matching the observations more closely to get a higher weight assigned, which indicates a higher probability of being in the true state

according to the Bayes theorem

$$P(p|z) = \frac{P(z|p)P(p)}{P(z)} \quad (5)$$

where p is the particle and z the measurement. After the weight updating step, low-weight particles are discarded, and new particles around the surviving particles are sampled using a resampling method. These aim to allocate new particles to each surviving particle based on its weight, ensuring the distribution accurately represents the underlying weights and a constant number of particles. The newly generated particles are then randomly moved away from their original position by an amount d to enable the exploration of new positions. Initially, d can be large enough for flexible particle movement. After every iteration, d gets smaller to hone into the final position. Popular resampling methods are systematic resampling, stratified resampling and residual resampling.

Finally, the current state μ is estimated by determining the weighted mean

$$\mu = \frac{1}{N} \sum_{i=1}^N p_i w_i \quad (6)$$

where p_i and w_i denote the i -th particle and the i -th weight, respectively.

This process works well for multiple iterations through the above steps. For example, one can use this to track a plane's position flying over a mountainous area and determine its position by measuring the height above ground by radar. First, all particles are distributed randomly over the area on the map, where the plane approximately is. Next, movement direction and speed can be put into the particle model by reading the compass and airspeed indicator, leading to an update of the particle position according to step two. The real-world height measurement, e.g., with radar, will be used as a third step to update the weights. Lastly, weight update, discarding, and resampling are done, and the position estimate is extracted from the weighted particle center. In this example, one can easily see that a single iteration may not yield a precise result. However, the moving plane collects a height profile during its movement, and these measurements are accumulated in the weighting of particles, leading to convergence around the actual position of the aircraft. Nevertheless, one can easily see that using a PF is unfeasible when moving over an ocean or a desert due to the low variance of measurement data. Consequently, the choice of variables is of high importance.

1) *Particle Filters for Probe Positioning:* For the PF method, the reference heatmap is interpolated to enable predictions of position and angle simultaneously with a single run.

The process of determining a position with Particle Filter (PF) is depicted in Fig. 4. Initially, the weights of all particles are equal and the particles are randomly distributed as shown in Fig. 4a. Since no movement by the template is present, step two is omitted, and the particles will not be moved actively. Next, each particle's weight is updated according to the differences between I and T when T is placed at the center

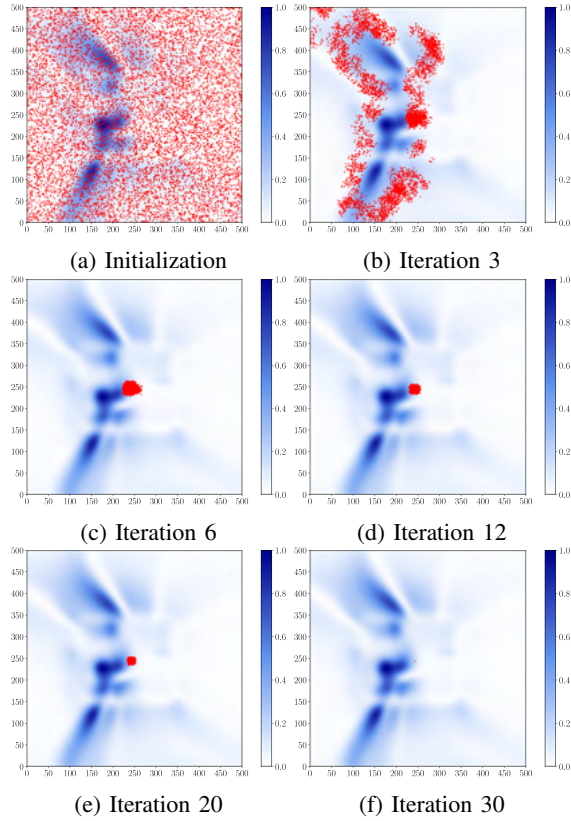


Fig. 4: Process of particle updating and finding the position of T in I for PF for 30 iterations. In (a) the red particles are distributed randomly and in the final step in (f) the position is determined precisely, since the particles converged into one point.

of gravity of the particles. There are many ways to obtain the difference between two heatmaps, where we use Mean Squared Error (MSE) as metric. With this information, the weights of particles are updated, and the resampling step takes place. This iterative updates the positioning of the particles and is shown in steps (b)-(f) in Fig. 4. Finally, we end up with an updated estimate closer to the actual position again at the center of gravity of the particles as Fig. 4f shows. Particles will eventually converge towards the actual position after a few iterations, which marks the end of classification. The accuracy of estimation is increased by increasing the number of iterations. The stability and convergence speed of the system is firmly bound to the initial coverage of the particles and thus proportional to the number of particles. More particles make positioning more accurate but also increase the computation effort.

IV. RESULTS

A. Measurement Setup

The measurement setup based on the NewAE Technology CW308 board is shown in Fig. 5 and the close-up view from the camera can be seen in Fig. 5b. We use STM32F303RCT7 microcontroller targets for the experiments,

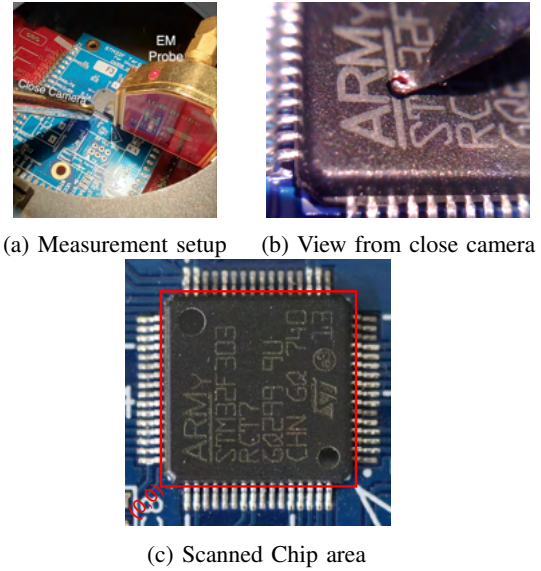


Fig. 5: Measurement setup with the STM32F303 on the blue PCB and the local EM-probe in red mounted on the x-y-z-table hovering closely above the microcontroller in (a). The detailed position can be determined by the silver camera reaching in from the left and a sample image from this camera can be seen in (b). The scanned chip area is shown in (c), while the bottom left point is the center of the coordinate system $S = (0,0)$.

running at 48 MHz. The setup can be freely moved through the Langer ICS 105 x-y-z-table, allowing EM probes to reach each position above the corner. The x-y-z-table features a positioning accuracy of $10\mu\text{m}$ for all three coordinate directions.

We capture separate heatmaps of EM emanations during the execution of the code in Listing 1 using two near-field probes⁵

- Langer ICR HH250-75 with an resolution of $150\mu\text{m}$ (referred to as EM150)
- Langer ICR HH150-75 with an resolution of $100\mu\text{m}$ (referred to as EM100)

while their signal is amplified by 30 dB using a Langer PA-303 low-noise amplifier and recorded by a PicoScope 6402D. The PicoScope is configured to a sampling rate of 2 GHz and a full-scale measurement range of 1 V. Since our approach aims to refine a position in a profiled attack scenario, we collect heatmap data for two different copies of the STM32F303RCT7 microcontroller. We denote one profiling chip and the other attack chip. We purposely use the very simple and trivial code in Listing 1 to emphasize the independence of any leakage prone implementations.

```

1 set_trigger();
2 volatile uint32_t foo = 0;
3 volatile uint32_t bar = 0xffffffff;
4
5 for (foo = 0; foo < 100000; foo++) {
6     bar ^= 0xffffffff;
7 }

```

⁵The resolution is based on the data-sheet published by Langer EMV-Technik GmbH.

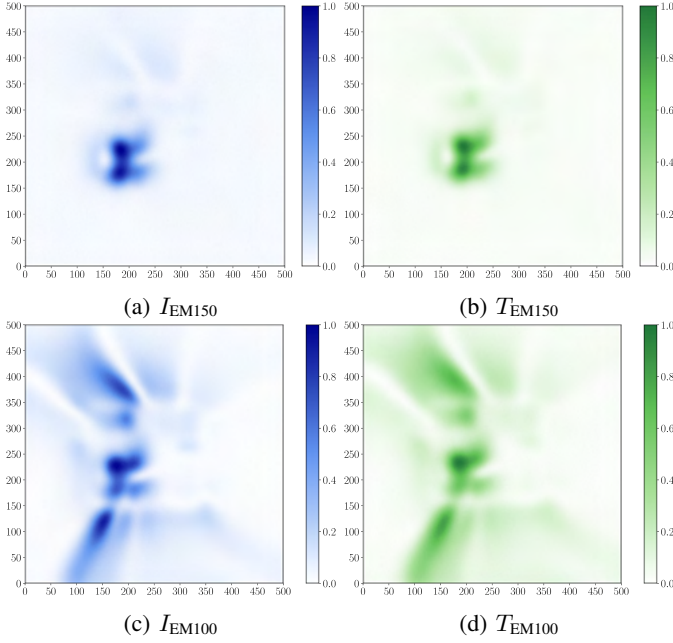


Fig. 6: Profiling I (blue) and attack heatmaps T (green) for both EM-probes EM150 (a), (b) and EM100 (c), (d) showing the variance at measurement position for profiling and attack chip.

```
8 reset_trigger();
```

Listing 1: Code under test.

We stepped over the area marked in Fig. 5c (10 mm by 10 mm) for generating the profiling heatmap I with a step width of $20\mu\text{m}$, giving us 501 positions in the x-direction and 501 positions in the y-direction with overall about 251000 positions covering an area of 1 cm by 1 cm. In all plots, the coordinate (0,0) is equal to the left bottom corner of the chip as marked in Fig. 5c. We plot the heatmap for the variance within one trace t at a single position calculated by

$$\text{Var}(t) = E[(t - \mu_t)^2] \quad (7)$$

where μ_t is the mean of t and E represents the expected value. Fig. 6 depicts the heatmaps for both probes once with the profiling device and once with the attacked microcontroller. Here, the heatmaps of the attack chip are purposely called T , since they cover as many points as I . The cut-outs T , which are needed for PROBENAV are a subset of T .

Our setup compares to Richter et al. very well on purpose, since we use the identical near-H-field probe EM100. Additionally, we use the EM150 to get more information on the robustness of our methods. Our x-y-z-table features a little higher precision of $10\mu\text{m}$ (in contrast to $20\mu\text{m}$ as Richter et al. estimate). However, to compare our results to Richter et al. we also scan our profiling map with $20\mu\text{m}$, so there is no difference in that regard.

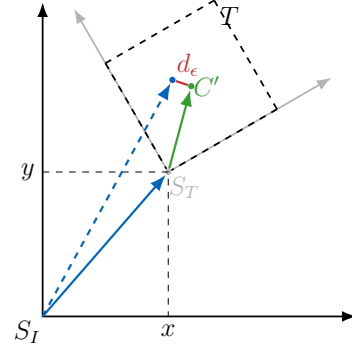


Fig. 7: Illustration of the error during reposition

B. Experiment Description

To allow for a consistent comparison between the four methods, every method is tasked with the same set of heatmaps. Fig. 6 shows the input maps I_{EM150} and I_{EM100} originating from the profiling chip for the subsequent test runs. The used probe is denoted by the subscript. For (re)positioning the heatmap T_{EM150} is used, which is a cut-out from T_{EM150} when measuring the attack chip with the EM150 probe. We omit the coordinates where T is cut from T to keep the naming short and since our evaluations picture the accuracy result for any possible cut-out T anyways. Due to the differences between the two probes, we do not test the transferability between probes with different resolution.

In order to save time, an attacker will aim to measure as little positions as possible while still being able to accurately place the probe on the attack chip. For testing the limits of our positioning methods, we cut-out smaller test maps of size 50×50 , 100×100 , 150×150 and 200×200 pixels from the respective T and test the accuracy for any possible cut out. This provides us with a set of about 90,000 to 203,400 experiments depending on the size of T for each probe and evaluation algorithm (CC, ORB, PF). In the following, we first summarize the results for each of the four methods separately and conclude the section with a comparison and recommendation.

C. Experimental Results

In the following, the results of all experiments with all four positioning methods described. Thereby, the distance between detected position C' and true position C of the center of T is calculated as

$$d_\epsilon = \sqrt{(x_C - x'_C)^2 + (y_C - y'_C)^2}. \quad (8)$$

The error is graphically illustrated in Fig. 7.

The rotation is not explicitly listed in the error equation, since the center coordinates x'_C, y'_C are calculated based on the positioning result, which is the bottom left corner S_T of the square T . Thus, the center calculation needs to take the rotation φ into account.

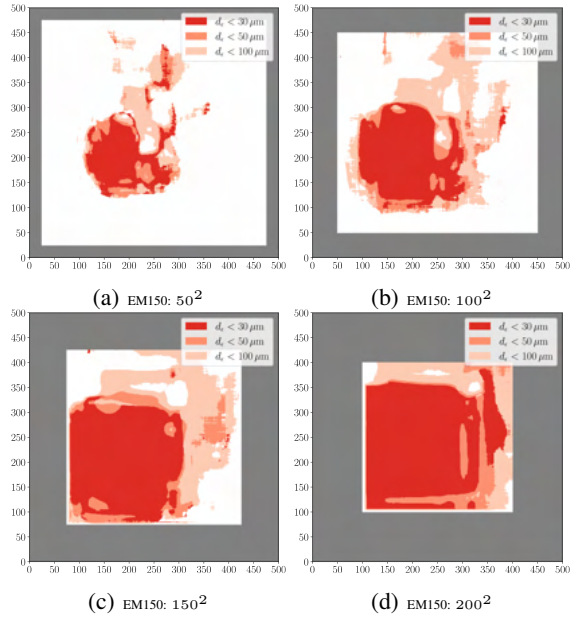


Fig. 8: Probe-positioning results by using CC with probe EM150. The plots (a)-(d) shows position accuracy increasing cut-out size. Center positions of T which can be classified with an distance d_e less then $100\mu\text{m}$, $50\mu\text{m}$, $30\mu\text{m}$ are colored in red with increasing saturation respectively. Gray areas show positions, which are not possible with the corresponding cut-out size.

1) *Correlation Coefficient*: The results for the experiments with CC as the detection algorithm are depicted in Fig. 8 for probes EM150 and EM100, respectively. The coloring is the same for Fig. 8 to Fig. 14. Center positions of T which can be classified with a distance d_e less than $100\mu\text{m}$, $50\mu\text{m}$, $30\mu\text{m}$ are colored in red with increasing saturation respectively. Gray areas show positions, which are not possible with the corresponding cut-out size, while white areas indicate center positions with an error $\geq 100\mu\text{m}$.

With the decreasing size of the cut-out, the error d_e increases, which is to be expected. For example, with the EM150-probe, about 61.8% of all positions for a 200^2 sized T result in an accuracy of at least $30\mu\text{m}$, while for a size of 50^2 the amount of positions decreases to 8.19%. The same behavior is noticed for the probe EM100, with 44.9% for 200^2 and 7.81% for 50^2 . The heatmaps with EM150 yield even better results for larger cut-outs T with a size of 150^2 or above compared to the EM100 probe.

For this method, we can conclude that in our experiments, both 150^2 and 200^2 give a high positioning accuracy for the both probes. Reliable positioning is possible when picking T from an area where most distinct features of the heatmap (around coordinates (200, 200)) are located. The results of CC and all other methods are summarized in Table I.

2) *ORB*: In Fig. 10 and Fig. 11 results for ORB are pictured with the same color encoding as before again for both probes. The discrete look of the error heatmaps for ORB results from

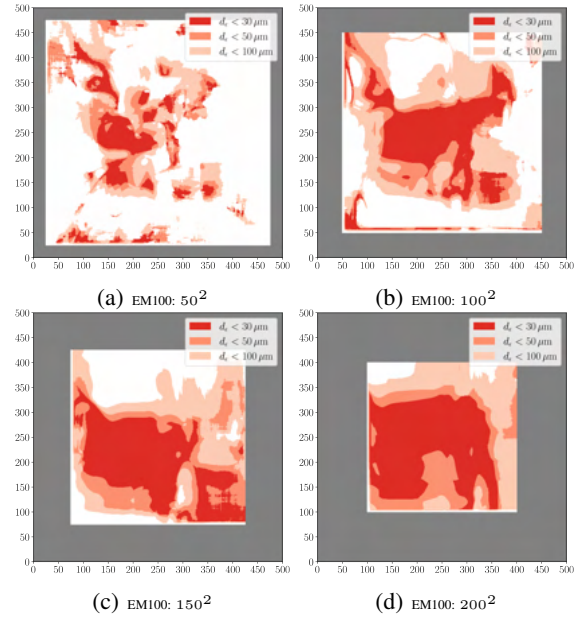


Fig. 9: Probe-positioning results by using CC with probe EM100 with increasing cut-out size from (a)-(d).

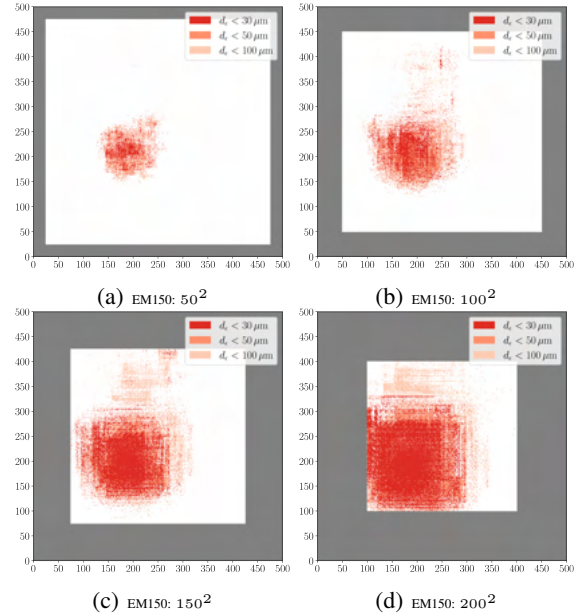


Fig. 10: Probe-positioning results by using ORB with decreasing cut-out size from (a)-(d) for probe EM150.

the classification algorithm, where the classification error can change from one pixel to the next. Concretely, due to the fact that the keypoints determining the features of the matches always have the same distance, also the matches are taken at these keypoints. If the cut-out T does not include the relevant features at the keypoints, the classification suffers. However, this effect can be reduced with larger cut-out sizes, as both Fig. 10d and Fig. 11d show.

We can again observe the same trend as in the CC results,

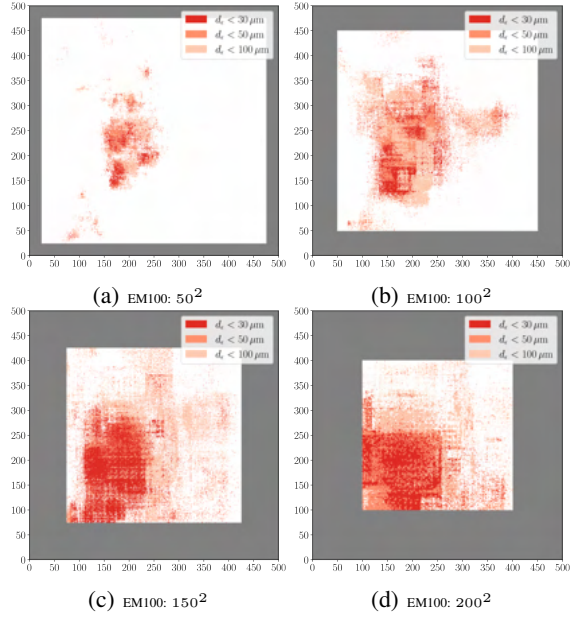


Fig. 11: Probe-positioning results by using ORB with decreasing cut-out size from (a)-(d) for probe EM100.

where EM150 performs significantly better than EM100. For larger sizes of T , EM100 and EM150 are approximately equal for ORB, but EM150 still yields better results similar to CC. This time, the difference is even higher, e.g., for T of size 200^2 in the case of the EM150 probe, 29.88% of all positions yield an error $d_e < 30\mu\text{m}$ while for EM100 only 12.27% can support an accurate probe positioning. Again, we can see that picking a cut-out at the area of most distinct features based on visual inspection will yield a consistently high accuracy for T of a size of at least 150^2 but precision increases for a size of 200^2 . For smaller cut-out sizes, the amount of high-accuracy positions degrades drastically.

3) *Particle Filter*: We use 10,000 particles in each run for both particle filter methods to identify the dependencies on the cut-out location. More particles can increase the positioning accuracy but also increase the computational effort. For the PF, we can again see increasing cut-out size having a beneficial effect on the positioning accuracy of most positions as Fig. 12 and Fig. 13 illustrate. Once again, the EM150 outperforms EM100. For the 200^2 cut-out, about 15.33% of all positions maintain a low error of $30\mu\text{m}$ for probe EM150, while for probe EM100, 9.49% of all positions have this level of accuracy. Again, picking a cut-out around the area with most features at position (200, 200) gives an accurate positioning result for size 150^2 , while a T of 200^2 does not bring a large improvement here. The small cut-out size of 50^2 primarily results in a high error positioning and, therefore, is not recommendable here. We conclude that PF is a reliable method for positioning errors lower than $50\mu\text{m}$ in our experiments with cut-out sizes of 150^2 and above.

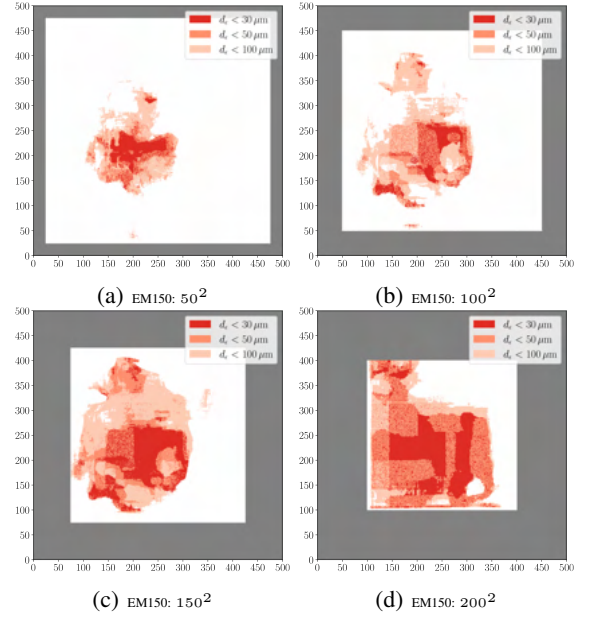


Fig. 12: Probe-positioning results by using PF with increasing cut-out size for (a)-(d) for the probe EM150.

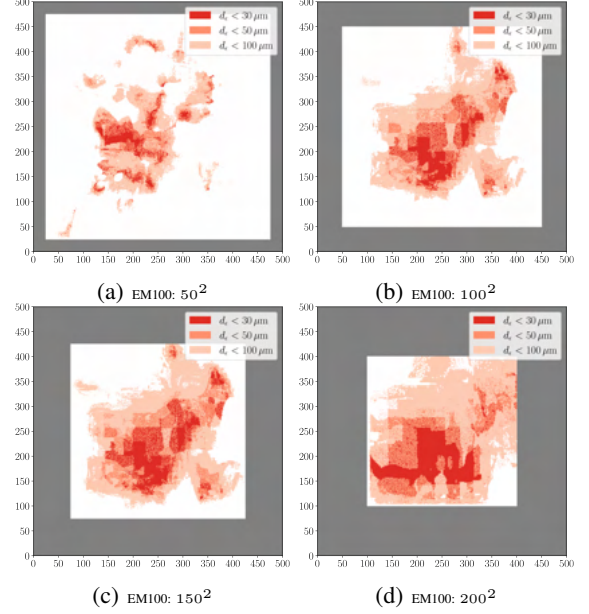


Fig. 13: Probe-positioning results by using PF with EM100 with increasing cut-out sizes from (a)-(d).

V. COMPARISON AND DISCUSSION

In summary, probe positioning is more accurate to manage when the positioning technique yields high accuracy independent of the choice of where to cut out T . Thus, the more dark red points are in the result maps, the better. Further, a relation of red points to features, which can be easily identified in the heatmap by visual inspection, is desirable as the second best option.

Our results for PROBENAV are shown in Table I and we

		50 ²			100 ²			150 ²			200 ²		
		CC	ORB	PF	CC	ORB	PF	CC	ORB	PF	CC	ORB	PF
EM150	30 μm	8.19	1.32	2.41	19.07	4.40	4.03	37.17	8.18	12.18	61.82	29.88	15.33
	50 μm	16.2	2.78	6.29	25.06	7.42	9.34	46.80	10.79	23.19	74.39	37.70	25.74
	100 μm	35.3	5.64	17.15	44.12	11.26	21.78	76.72	16.35	42.39	95.12	49.40	31.76
EM100	30 μm	7.81	1.31	2.82	20.20	4.89	6.02	30.83	9.73	9.41	44.91	12.27	9.49
	50 μm	10.92	5.34	2.30	32.96	10.94	14.90	46.80	14.08	22.55	65.43	17.56	18.90
	100 μm	18.16	10.39	3.80	60.75	22.81	35.47	76.25	25.52	51.14	94.20	28.71	40.34

TABLE I: Accuracy results for all methods. Methods with high accuracy at feature rich areas are marked bold.

summarize a few observations here. Independent of the EM-probe, a bigger cut-out has a beneficial influence on the number of high-accuracy points. CC and ORB perform reliably well for T of size 200² and are robust for that size when picking an area of the heatmap with visible features. Oriented FAST and rotated BRIEF (ORB) gains the most accuracy from an increase of the cut-out size. CC is still better in any experiment. PF features less areas with high accuracy and is therefore not a reliable positioning method in our experiments. The suitable techniques with high precision ($d_e < 30 \mu\text{m}$) and window-sizes are marked in bold in Table I, i.e., where overall accuracy is high and where a pick based on visual features is possible based on our results. The required area can easily be determined by human vision, since one just needs to pick any T roughly in the center of the chip. Thus, ORB and most significantly CC provide robust and accurate positioning with an error of $d_e < 30 \mu\text{m}$. Especially CC shows the highest accuracy of all methods at all cut-out sizes and both probes and is therefore our recommend positioning metric. In Fig. 14, we showcase the areas an adversary needs to be able to find by visual inspection for achieving peak accuracy. Concretely, the center of the 200² point square needs to be within the **blue** rectangle for EM100 and in the **green** rectangle for EM100 in Fig. 14c. For large sizes of T of at least 150², CC is mostly independent of the choice of T as long as it is in the middle of the package. Thus, it is our recommend choice of positioning algorithm.

Regarding the positioning duration, collecting the required 22,500 traces for the 150² cut-out T is highly dependent on the movement speed of the x-y-z table. In our case, this takes about 5 hours for the 150² positions, while the calculation of the position with the named methods is about half a minute at maximum⁶. However, collecting the initial heatmap I with 501x501 position will take about 2d. Remember that collecting the profiling map I is necessary once for every chip model. T is necessary for every (re)positioning and can be managed in a reasonable amount of time.

Compared to Richter et al. [9], who claim an average accuracy of around 30 μm , PROBENAV can predict the probe position with an average error less than 30 μm . However, compared to their approach, we do not require large datasets, point-of-interest, and machine learning to precisely reposition a EM-near-field probe. Furthermore, Richter et al. need to collect 200 traces per position, which takes little less than

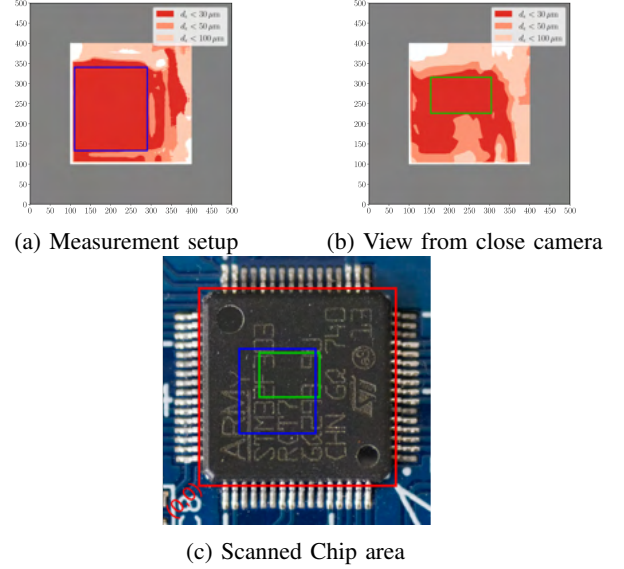


Fig. 14: We put the high accuracy areas in respect to the chip surface in order to showcase the ease of positioning. For that, we choose a rectangle, that an adversary needs to position the center of the 200 by 200 position square in, in order to achieve a repositioning accuracy of 30 μm or higher. The rectangles for the EM100 probe is marked in **blue** in (a) and for EM100 in **green** in (b). The areas are also added to the image of the microcontroller in (c).

11.6d to record all traces for the same area coverage as we have⁷. Thus, for profiling PROBENAV provides a speed-up of a factor of 2.9 while every repositioning will have an advantage of a factor of 27.9. Further, also a lower SNR is sufficient for precise positioning since we evidence the feasibility at a non-decapped chip. We estimate that the methodology by Richter et al. will work similarly in our scenario. However, more traces may be required to compensate for the lower SNR. While PROBENAV requires to save a single image of the size of the profiling heatmap I , Richter et al. will store the trained neural network. Thus, both methods do not have large differences in that regard.

Overall, ProbeNav exceeds the state-of-the-art in terms of profiling and classification speed while keeping the same accuracy. The top performance in our test was observed with

⁶Duration of methods for one positioning run: CC=0.04 s; ORB=7.8 s; PF=38 s.

⁷Capturing traces for one device takes 5.81d (11.6d for profiling and attack device) when assuming a capture rate of 100 traces per second and the same movement time as for our table.

the simplest positioning technique - CC - which is also well explainable.

VI. CONCLUSION

In this work, we present our methodology called PROBE-NAV for accurate repositioning a local EM probe. Our method allows the usage of different localization methods, where we tested CC, ORB and PF. CC performs best at our example of two STM32F303 microcontroller in a profiled attack scenario and we achieve a maximal error for these methods of 30 μm at a feature rich area in the heatmap, which can be identified easily by visual inspection. Thus, we recommend the usage of PROBE-NAV with CC based on our results. We also showed our methods to work on little data and being about 3 times faster for profiling and 28 faster for repositioning compared to state-of-the-art techniques, due to requiring only a single trace per position. **The implementation of PROBE-NAV including the three methods PF, ORB and CC will be open sourced on gitlab.**

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